Notes
Mainly to Elements of Set Theory

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$B \subseteq C \implies \mathscr{P}(B) \subseteq \mathscr{P}(C)$

Proof. By def $\mathscr{P}(B) = \{x \mid x \subseteq B\}$
$\forall x, x \in \mathscr{P}(B)$, Then, $x \subseteq B \subseteq C$
$\implies x \subseteq C$
$\implies x \in \mathscr{P}(C)$ By def $\mathscr{P}C = \{x \mid x \subseteq C\}$
Thus, $\mathscr{P}(B) \subseteq \mathscr{P}(C)$

$\square$

A is a finite set,$|A| = n \implies |\mathscr{P}(A)| = 2^n$

Proof. Consider taking i elements from A to build a subset.
Then $i \in \{0, \ldots, n\}$, and the number of that kind of subsets is $\binom{n}{i}$.
(if $i = 0$, the subset is $\emptyset$; if $i = n$, the subset is A itself)
$\implies |\mathscr{P}(A)| = \sum_{i=0}^{n} \binom{n}{i}$
$\implies |\mathscr{P}(A)| = 2^n$, By Binomial Theorem, consider $(1+1)^n$

$\square$

$x, y \in B \implies \{\{x\}, \{x, y\}\} \in \mathscr{P}(\mathscr{P}(B))$

Proof. $x, y \in B$
$\implies \{x\}, \{x, y\} \in \mathscr{P}(B)$
$\implies \{\{x\}, \{x, y\}\} \subseteq \mathscr{P}(B)$
Thus, by def $\{\{x\}, \{x, y\}\} \in \mathscr{P}(\mathscr{P}(B))$ $\square$

Remark. *A simple way to calculate the rank of set (which finite) is to count the deepest nested brackets until meet atom or $\emptyset$.*

$\forall \alpha \in \mathbb{N}, \alpha \leq 3, V_{\alpha+1} = A \cup \mathscr{P}(V_\alpha)$

Proof. $V_0 = A$
By def, $V_{n+1} = V_n \cup \mathscr{P}(V_n)$
$V_1 = V_0 \cup \mathscr{P}(V_0) = A \cup \mathscr{P}(V_0)$
$V_2 = V_1 \cup \mathscr{P}(V_1) = A \cup \mathscr{P}(V_0) \cup \mathscr{P}(V_1) = A \cup \mathscr{P}(V_1)$
$\mathscr{P}(V_0) \cup \mathscr{P}(V_1) = \mathscr{P}(V_1)$, By $\mathscr{P}(V0) \subseteq \mathscr{P}(V_1)$ since $V_0 \subseteq V_1$, by previous result.
$V_3 = V_2 \cup \mathscr{P}(V_2) = A \cup \mathscr{P}(V_1) \cup \mathscr{P}(V_2)$ Similarly,
$V_3 = A \cup \mathscr{P}(V_2)$ $\square$

Remark. *I guess we can use mathematical induction to make it to $\forall \alpha \in \mathbb{N}$*

Remark. *All sets are classes, $\{x \mid P(x)\}$ is appropriately is class-builder, where $P(x)$ is open formula.*

Axiom 1. *Extensionality Axiom*

$$\forall A \forall B, (\forall x, x \in A \iff x \in B) \implies A = B$$

Axiom 2. *Empty Set Axiom*

$$\exists A \forall x, x \notin A$$

Remark. *Does $\emptyset$ act like a unique generator in terms of abstract algebra? (e.g. similar as prime numbers can be used to generate whole $\mathbb{N}$ under multiplication)*

Axiom 3. *Pairing Axiom*

$$\forall A \forall B \exists C \forall x (x \in C \iff x = A \lor x = B)$$

Axiom 4. *Union Axiom*

Axiom 5. *Power Set Axiom*

$$\forall A \exists B \forall x (x \in B \iff x \subseteq A)$$

Axiom 6. *Subset Axiom*

Axiom 7. *Replacement Axiom*

Axiom 8. *Infinity Axiom*

Axiom 9. *Regularity Axiom*

Axiom 10. *Choice Axiom*

Remark. *Definition is like an abbreviation of syntax, and it seems that it must be iff statements.*